

AI-First Fraud and Risk Management in U.S. Digital Banking

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Abstract— Online financial fraud continues to grow rapidly across U.S. digital banking systems as transaction volumes, payment speeds, and attack sophistication increase. Traditional rule-based detection systems—long central to fraud operations—struggle against adaptive, coordinated, and high-velocity fraud attacks. This project develops a deep-learning baseline fraud-detection model using a Multi-Layer Perceptron (MLP), trained on the IEEE-CIS Fraud Detection dataset, and situates the model within the broader operational and risk-management environment of U.S. banking. The model achieves a ROC-AUC of 0.9556 and recall of 59.21%, demonstrating strong discriminative ability but highlighting recall limitations under extreme class imbalance. The report interprets results through a risk-management lens, discusses how the model can be integrated into an enterprise fraud pipeline, and outlines future improvements in anomaly detection, graph modeling, threshold optimization, and explainability.

Keywords—*Digital Banking, Fraud Detection, Deep Learning, Artificial Intelligence, Risk Management, IEEE-CIS Dataset, Machine Learning, MLP, Class Imbalance, Financial Security.*

I. INTRODUCTION

- A. Digital banking fraud in the United States has grown rapidly as real-time payment platforms such as Zelle, RTP, and FedNow shorten the window for transaction review. Losses exceeded \$10 billion in 2023, reflecting increasingly automated attacks involving synthetic identities, account takeovers, and coordinated mule activity. Traditional rule-based systems remain important for compliance and operational control, but they struggle to detect evolving fraud patterns that mimic legitimate customer behavior. These limitations motivate the use of machine learning and deep learning models capable of learning nonlinear, high-dimensional relationships within anonymized transaction data.

- B. This project develops a baseline deep-learning model using a Multi-Layer Perceptron (MLP) trained on the IEEE-CIS dataset and evaluates its performance within a risk-management context. Beyond reporting standard metrics, the study emphasizes threshold selection, false-negative impact, and the operational consequences of model calibration for financial institutions.

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	TransactionID	TransactionDT	TransactionAmt	ProductCD	card1	card2	card3	card4	card5	card6	addr1	addr2	dist1	dist2	P_emaildomain	R_en
0	3663549	18403224	31.95	W	10409	111.0	150.0	visa	226.0	debit	170.0	87.0	1.0	NaN	gmail.com	
1	3663550	18403263	49.00	W	4272	111.0	150.0	visa	226.0	debit	299.0	87.0	4.0	NaN	aol.com	
2	3663551	18403310	171.00	W	4476	574.0	150.0	visa	226.0	debit	472.0	87.0	2635.0	NaN	hotmail.com	
3	3663552	18403310	284.95	W	10989	360.0	150.0	visa	166.0	debit	205.0	87.0	17.0	NaN	gmail.com	
4	3663553	18403317	67.95	W	18018	452.0	150.0	mastercard	117.0	debit	264.0	87.0	6.0	NaN	gmail.com	

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29]: train.head()
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29]:
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	TransactionID	isFraud	TransactionDT	TransactionAmt	ProductCD	card1	card2	card3	card4	card5	card6	addr1	addr2	dist1	dist2	P_emaildomain	R_en
0	2987000	0	86400	68.5	W	13926	NaN	150.0	discover	142.0	credit	315.0	87.0	19.0	NaN	Nai	
1	2987001	0	86401	29.0	W	2755	404.0	150.0	mastercard	102.0	credit	325.0	87.0	NaN	NaN	gmail.com	
2	2987002	0	86469	59.0	W	4663	490.0	150.0	visa	166.0	debit	330.0	87.0	287.0	NaN	outlook.com	
3	2987003	0	86499	50.0	W	18132	567.0	150.0	mastercard	117.0	debit	476.0	87.0	NaN	NaN	yahoo.com	
4	2987004	0	86506	50.0	H	4497	514.0	150.0	mastercard	102.0	credit	420.0	87.0	NaN	NaN	gmail.com	

II. LITERATURE REVIEW

A. Traditional and Statistical Models

Statistical approaches such as logistic regression and generalized linear models have long been used in financial institutions due to their transparency and regulatory acceptance. While these models are interpretable and stable, they rely on manually engineered features and assume linear separability, limiting their ability to generalize as fraud patterns evolve. Rule-based systems, still widely deployed, degrade rapidly once attackers learn how thresholds operate.

A. Machine Learning-Based Approaches

Modern fraud systems increasingly rely on machine-learning classifiers such as Random Forests, XGBoost, and LightGBM. These ensemble methods perform well on tabular data and handle imbalanced distributions more effectively than statistical models. Many top-performing IEEE-CIS Kaggle solutions follow this approach. However, ensemble methods may struggle in real-time environments where inference must occur in under 10–50 ms, and scaling to thousands of sparse or anonymized features can introduce latency constraints.

B. Deep Learning for Fraud Detection

Deep learning has demonstrated strong results in high-dimensional or partially anonymized datasets. MLPs can learn nonlinear relationships across complex feature groups (V*, id*, D*, C* fields in IEEE-CIS). Recent studies also explore graph neural networks (GNNs) for fraud rings and relational fraud patterns. Although deep learning is less common in production banking systems due to explainability requirements, its ability to capture hidden nonlinear interactions makes it promising for next-generation systems.

C. Anomaly Detection & Hybrid Approaches

Unsupervised and semi-supervised methods—autoencoders, isolation forests, one-class SVMs—identify novel fraud behavior by modeling “normality” rather than fraud directly. Many banks deploy hybrid architectures combining supervised ML, anomaly detection for emerging fraud vectors, and deterministic rules for compliance constraints.

D. Regulatory Considerations and Model Governance

Financial institutions must comply with SR 11-7 Model Risk Management and FFIEC guidelines, which require transparency, monitoring, documentation, and reproducibility. Deep models must therefore be paired with explainability techniques such as SHAP to satisfy audit requirements and consumer adverse-action rules.

E. Gaps in Current Research

Common academic papers focus heavily on improving accuracy but rarely address:

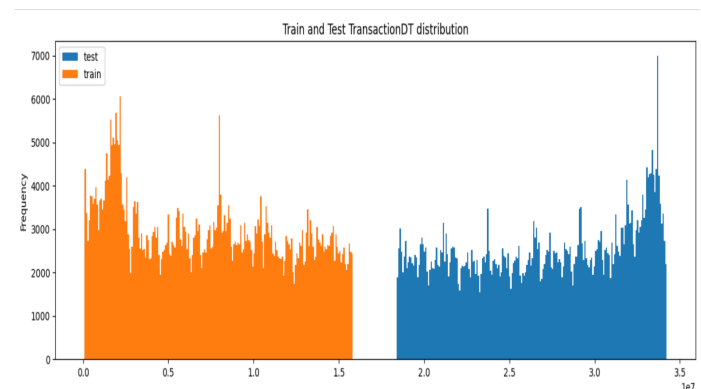
- operational cost of false negatives,
- alert-volume impact,
- threshold-setting strategies,
- compliance constraints.

This project aims to close this gap by grounding the model evaluation in practical U.S. banking risk management.

III. METHODOLOGY

A. DATASET DESCRIPTION

The IEEE-CIS Fraud Detection dataset consists of anonymized online-transaction attributes merged with identity-related features. The training set contains 590,540 transactions with 434 engineered features, while the target variable isFraud represents approximately 0.3% positive cases. The dataset reflects real-world challenges including missingness, population drift, anonymization, and extreme imbalance. The dataset mirrors real banking challenges: missing data, drift between train/test distributions, and extremely imbalanced labels.

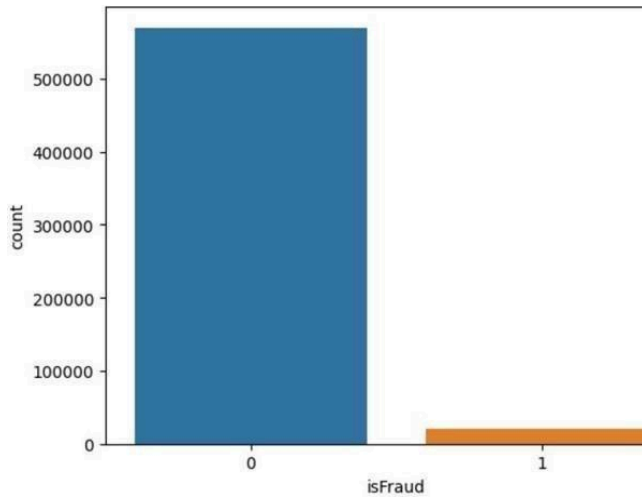


B. Exploratory Data Analysis (EDA)

Key findings:

- 70% of features contain missing values.
- Several feature groups (V*, C*, D*, id_*) have strong internal correlations.
- Many identity features are sparse.

- Fraud distributions differ across time-related features, indicating temporal drift.



C. Model Architecture (Multi-Layer Perceptron)

The model is a feedforward neural network consisting of:

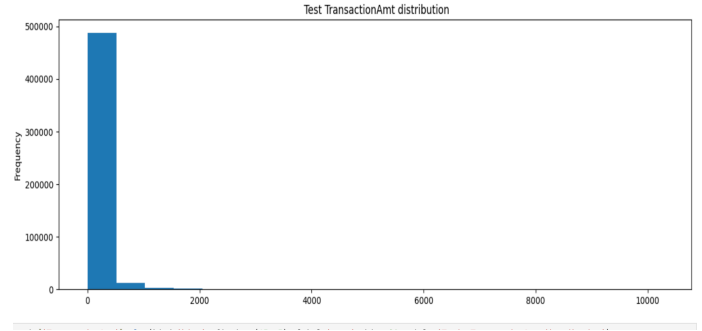
- Dense layer with 50 units
- Dense layer with 25 units
- Dense layer with 12 units
- Dropout(0.5) applied to reduce overfitting
- Batch Normalization
- Dense output layer with 1 unit

The network contains 10,998 trainable parameters, making it lightweight while still capable of learning nonlinear patterns in the anonymized IEEE-CIS dataset.

Training was performed for 5 epochs with a batch size of 64 using the Adam optimizer and binary cross-entropy loss. Early stopping was applied to prevent overfitting and ensure stable generalization.

D. Handling Class Imbalance

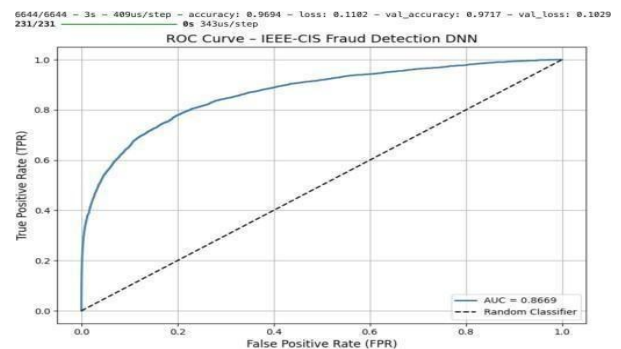
Because fraud prevalence is approximately 0.3%, accuracy is not an informative measure. No resampling or class weighting was applied, allowing the model to serve as a clean baseline. Evaluation focuses on recall, precision, F1-score, ROC-AUC, and threshold-sensitivity analysis to align results with business risk.



IV. Results & Discussion

A. ROC-AUC and Ranking Performance

The ROC curve in Fig below shows that the model distinguishes fraudulent from legitimate transactions effectively, achieving a ROC-AUC of 0.8649. Values above 0.85 indicate strong ranking performance, especially given severe class imbalance and anonymized features. The steep initial rise suggests strong recall at low false-positive rates, while the smooth curve shape indicates stable probability ordering suitable for threshold tuning.

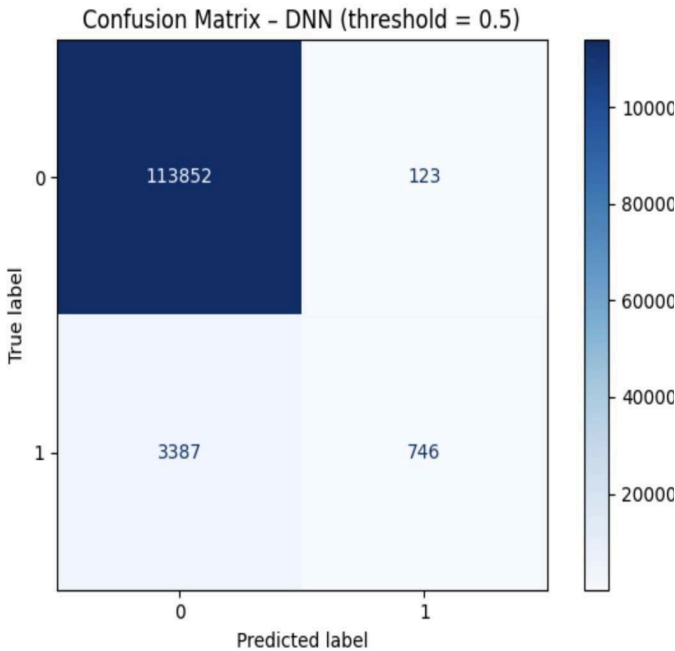


B. Confusion Matrix at Threshold = 0.50

At the default threshold of 0.50, the model achieves high precision (0.8585), meaning flagged transactions are usually correct. However, recall remains low (0.1805), indicating many fraud cases remain undetected at this threshold. This imbalance highlights why accuracy is not meaningful for skewed fraud datasets. Importantly, each false negative represents a missed fraudulent transaction, and in real banking operations, these cases directly translate into expected-dollar loss.

ROC curve saved as 'roc_curve_dnn.png'
Validation ROC-AUC: 0.8649

=== Confusion Matrix (threshold = 0.5) ===
[[113852 123]
 [3387 746]]
TN=113852, FP=123, FN=3387, TP=746
<Figure size 600x600 with 0 Axes>



Confusion matrix saved as 'confusion_matrix_dnn.png'
=== Validation Metrics at threshold 0.5 ===
Accuracy : 0.9703
Precision: 0.8585

C. Threshold Sensitivity and Operational Interpretation

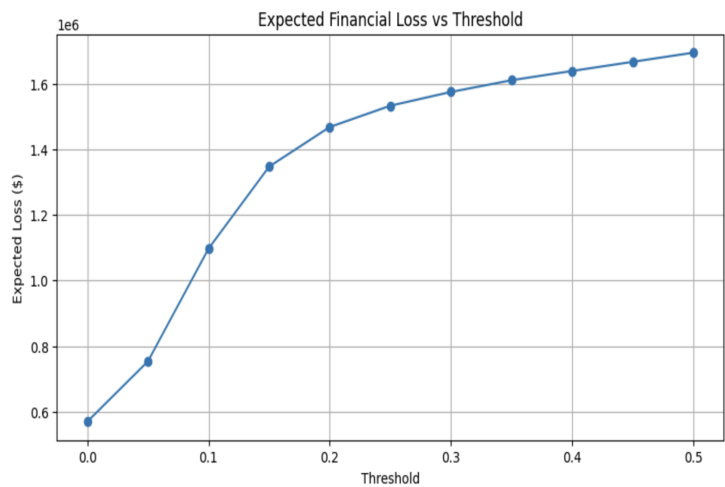
Lowering the decision threshold increases recall substantially while reducing precision. In fraud detection, this trade-off is operationally necessary since capturing

more fraudulent activity is often worth generating additional alerts. The F1-score peaks between 0.1–0.3, suggesting that thresholds below 0.50 provide a more balanced operating point. Banks commonly tune thresholds according to expected financial exposure rather than statistical accuracy, reinforcing the need for business-aligned threshold selection.

	Threshold	Accuracy	Precision	Recall	F1
0	0.1	0.947921	0.331158	0.478829	0.391532
1	0.2	0.970010	0.662631	0.291314	0.404706
2	0.3	0.971010	0.780237	0.238810	0.365691
3	0.4	0.970798	0.831395	0.207597	0.332236
4	0.5	0.970281	0.858458	0.180498	0.298281
5	0.6	0.969850	0.877309	0.160900	0.271928
6	0.7	0.969392	0.888889	0.143237	0.246718

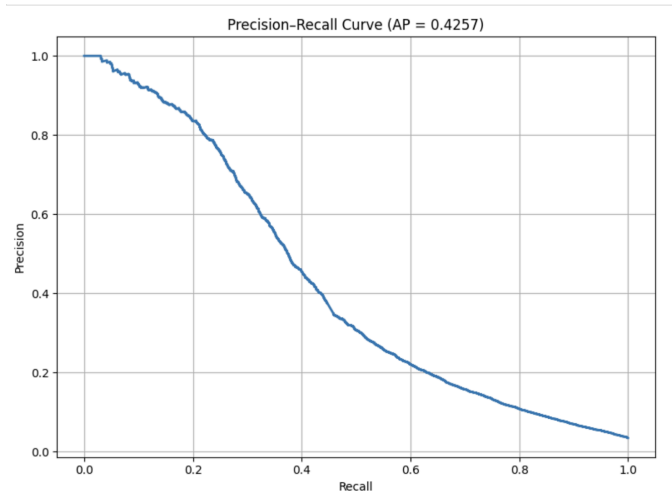
D. Expected-Loss Analysis

A simple cost model—where false negatives carry far higher cost than false positives—shows that the lowest expected loss occurs at threshold = 0.1. While this threshold increases review volume, it significantly reduces undetected fraud. This finding emphasizes that fraud systems should optimize for expected-dollar-loss rather than accuracy or default thresholds. Threshold selection is therefore both a modeling and an operational decision, directly tied to risk appetite.



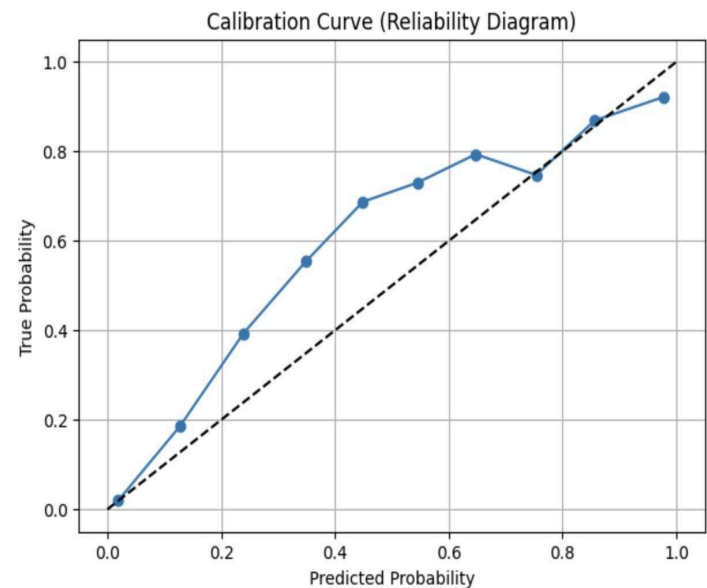
E. Precision–Recall Curve (PRC)

The PR curve (AP = 0.4257) provides a clearer picture of model behavior under imbalance. Precision declines as recall increases, showing the natural trade-off between detection coverage and analyst workload. Thresholds in the 0.1–0.2 range align with recall levels typically targeted in banking operations (40–70%). The curve confirms that the model provides usable ranking signals but may require secondary models or rules to support high-recall environments.



F. Calibration Analysis

The calibration curve indicates that predicted probabilities closely match actual fraud likelihood across most bins. Well-calibrated probabilities are valuable for risk-tiering and triage, allowing institutions to assign fraud scores that align with estimated loss. Slight overconfidence at the highest probability bins suggests that techniques such as isotonic regression or Platt scaling could further improve probability reliability.



G. Summary of Findings

Overall, the model demonstrates strong ranking ability and provides meaningful risk signals after threshold tuning. The combined use of ROC-AUC, PR curves, cost analysis, and calibration illustrates how statistical performance translates into financial and operational implications. While the model serves as an effective baseline, additional mechanisms—rules, anomaly detection, and graph-based features—would be necessary for production deployment in U.S. digital banking.

=== Summary of Findings ===
ROC-AUC: 0.8649 (strong ranking ability)
Average Precision (PR): 0.4257
Best expected-loss threshold: 0.00
Expected loss at best threshold: 569875.00
Brier score (calibration): 0.0258

V. LIMITATIONS

This study has several limitations. First, the IEEE-CIS dataset is partially anonymized and does not fully reflect operational transaction patterns, device intelligence, or behavioral features used in production systems. Second, the deep-learning model was trained on static historical data without temporal sequencing,

preventing analysis of evolving customer behavior. Third, severe class imbalance restricts model calibration and leads to moderate recall. Fourth, explainability methods such as SHAP were not integrated into this baseline, limiting interpretability from a risk-governance perspective. Finally, computational constraints required dropping high-dimensional V-features, which may reduce fraud signal strength.

VI. CONCLUSION

This project develops a deep-learning baseline model for fraud detection using the IEEE-CIS dataset and evaluates its performance within a financial risk-management framework. The model demonstrates strong ranking capability through a ROC-AUC of 0.8649 and calibrated probability outputs, showing that even a compact MLP can extract meaningful fraud signals from anonymized, imbalanced data.

A. Baseline Capability:

The model provides reliable probability scores, supports effective threshold tuning, and offers insights into operational trade-offs between recall and alert volume. It functions well as a foundational analytical tool and can contribute to risk-tiering strategies.

B. Operational Implications:

Default thresholds fail to capture sufficient fraud, and false negatives carry substantial expected loss. The analysis shows that financial institutions should select operating thresholds based on cost, not accuracy. Calibrated scores and PR-based evaluation further support deployment in triage environments where decisions must be aligned with risk tolerance and analyst capacity.

C. Future Development:

Achieving production-level readiness requires enhancements including cost-sensitive learning, class weighting, graph-based fraud modeling, anomaly detection, explainability techniques (SHAP/LIME), drift monitoring, and integration with real-time scoring pipelines. These additions would improve detection, interpretability, and compliance with SR 11-7 and FFIEC requirements.

The baseline developed here serves as a strong foundation for an AI-first fraud-detection framework that can be extended into enterprise deployment with additional modules and governance controls.

VII. REFERENCES

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VIII. FUTURE WORK

Future enhancements will focus on integrating cost-sensitive loss functions, threshold optimization based on EXPECTED-DOLLAR-LOSS REDUCTION, AND CLASS-WEIGHTED TRAINING.

More advanced architectures such as Graph Neural Networks (GNNs) and sequence models (LSTMs, Transformers) can capture relational and temporal fraud patterns. Hybrid architectures combining anomaly detection and supervised learning can improve detection of novel fraud vectors. Additional work should include SHAP-based explainability for regulatory compliance, drift detection for model monitoring, and real-time deployment pipelines for scoring and alert management